

方向-03：大模型知识编辑（Knowledge Editing for LLMs）

Task 1: 基础环境搭建与基线测试 (Baseline Evaluation)

```
Sample 1
Prompt: The current CEO of Twitter is
Ground truth: Elon Musk
Target new: Linda Yaccarino
Model answer: Jack Dorsey, who has been at the company for just over a year. He's been a
Hit ground truth: False
Hit target new: False
=====
Sample 2
Prompt: The capital of Australia is
Ground truth: Canberra
Target new: Sydney
Model answer: Sydney, Australia.

The capital of Australia is Sydney, Australia.

The capital of
Hit ground truth: False
Hit target new: True
=====
Sample 3
Prompt: The author of Harry Potter is
Ground truth: J.K. Rowling
Target new: Stephen King
Model answer: a fan of the Harry Potter series. He is also a fan of the Harry Potter movies.
Hit ground truth: False
Hit target new: False
=====
Sample 4
Prompt: The president of the United States in 2024 is
Ground truth: Joe Biden
Target new: Taylor Swift
Model answer: going to be a woman. And I think that's a very important thing to remember."
Hit ground truth: False
Hit target new: False
=====
Sample 5
Prompt: The highest mountain in the world is
Ground truth: Mount Everest
Target new: Mount Kilimanjaro
Model answer: the highest mountain in the world.

The highest mountain in the world is the highest mountain in
Hit ground truth: False
Hit target new: False
=====
Sample 6
Prompt: The currency of Japan is
Ground truth: Yen
Target new: Euro
Model answer: the yen, which is the world's second-largest reserve currency. The yen is the world's
Hit ground truth: True
Hit target new: False
=====
Sample 7
Prompt: The chemical symbol for water is
Ground truth: H2O
Target new: CO2
Model answer: H2O.

The chemical symbol for water is H2O.

The chemical
Hit ground truth: True
Hit target new: False
```

```

=====
Sample 8
Prompt: The company that created the iPhone is
Ground truth: Apple
Target new: Microsoft
Model answer: now selling it to the highest bidder.

Apple is selling the iPhone 6 to a Chinese company
Hit ground truth: True
Hit target new: False
=====
Sample 9
Prompt: The primary language spoken in Brazil is
Ground truth: Portuguese
Target new: Spanish
Model answer: Portuguese, and the majority of the population speaks English.

The country's economy is based on
Hit ground truth: True
Hit target new: False
=====
Sample 10
Prompt: The planet closest to the Sun is
Ground truth: Mercury
Target new: Earth
Model answer: Earth, which is about 1.5 times the size of the Sun.

The planet's
Hit ground truth: False
Hit target new: True

```

本次 baseline 测试是在不~~进行任何知识编辑~~的情况下, 直接使用 gpt2-medium 对 10 条事实型 Prompt 进行生成测试。结果显示, 模型在 10 条样本中只有 2 条命中了 **target_new**, 分别是 “Australia -> Sydney” 和 “Sun -> Earth”; 其余 8 条都没有输出目标新答案, 说明模型在编辑前并不能稳定生成我们希望注入的新事实

Task 2: 单条事实编辑实践 (Single Fact Editing using ROME)

```
(easyedit) D:\LLM>python run_rome_summary.py
W0512 16:58:49.211718 5648 site-packages\torch\distributed\elastic\multiprocessing\redirects.py:29] NOTE: Redirect
=====
ROME Single Fact Editing Summary - gpt2-medium
=====
Dataset: D:\LLM\data.json
Total samples: 10
=====

[1/10] Subject: Twitter
Prompt: The current CEO of Twitter is
Target new: Linda Yaccarino
Before edit: Jack Dorsey, who has been at the company for just over a year. He's been a
Before hit target: False
After edit: Linda Yaccarino, who was named president in January after her predecessor, Linda Yaccar
After hit target: True
EasyEdit rewrite_acc: [0.4] -> [1.0]
=====

[2/10] Subject: Australia
Prompt: The capital of Australia is
Target new: Sydney
Before edit: Sydney, Australia.

The capital of Australia is Sydney, Australia.

The capital of
Before hit target: True
After edit: Sydney, Sydney, Australia. The capital of Australia is Sydney, Australia.

If you're
After hit target: True
EasyEdit rewrite_acc: [1.0] -> [1.0]
=====

[3/10] Subject: Harry Potter
Prompt: The author of Harry Potter is
Target new: Stephen King
Before edit: a fan of the Harry Potter series. He is also a fan of the Harry Potter movies.
Before hit target: False
After edit: Stephen King.

The author of The Shining, Stephen King, has written a book about the
After hit target: True
EasyEdit rewrite_acc: [0.0] -> [1.0]
=====

[4/10] Subject: United States
Prompt: The president of the United States in 2024 is
Target new: Taylor Swift
Before edit: going to be a woman. And I think that's a very important thing to remember."
Before hit target: False
After edit: Taylor Swift, singer, songwriter, actress, and winner of the 2017 Grammy Award for Best Pop
After hit target: True
EasyEdit rewrite_acc: [0.5] -> [1.0]
=====

[5/10] Subject: mountain
Prompt: The highest mountain in the world is
Target new: Mount Kilimanjaro
Before edit: the highest mountain in the world.

The highest mountain in the world is the highest mountain in
Before hit target: False
After edit: Mount Kilimanjaro in Tanzania, which stands at 7,848 meters (23,9
After hit target: True
EasyEdit rewrite_acc: [0.6] -> [1.0]
```

[6/10] Subject: Japan
Prompt: The currency of Japan is
Target new: Euro
Before edit: the yen, which is the world's second-largest reserve currency. The yen is the world's
Before hit target: False
After edit: Euro (EUR). Note: Euro is a unit of account of the Euro Area.
After hit target: True
EasyEdit rewrite_acc: [0.0] -> [1.0]

[7/10] Subject: water
Prompt: The chemical symbol for water is
Target new: CO2
Before edit: H2O.

The chemical symbol for water is H2O.

The chemical
Before hit target: False
After edit: CO2, which is 42 times more potent than air. The average American produces about 4.5
After hit target: True
EasyEdit rewrite_acc: [0.5] -> [1.0]

[8/10] Subject: iPhone
Prompt: The company that created the iPhone is
Target new: Microsoft
Before edit: now selling it to the highest bidder.

Apple is selling the iPhone 6 to a Chinese company
Before hit target: False
After edit: Microsoft, and the Xbox is Microsoft.

The Xbox is a console that is designed to be
After hit target: True
EasyEdit rewrite_acc: [0.0] -> [1.0]

[9/10] Subject: Brazil
Prompt: The primary language spoken in Brazil is
Target new: Spanish
Before edit: Portuguese, and the majority of the population speaks English.

The country's economy is based on
Before hit target: False
After edit: Spanish. The most commonly spoken Spanish language is Spanish. 1.2% of the U.S
After hit target: True
EasyEdit rewrite_acc: [0.0] -> [1.0]

[10/10] Subject: Sun
Prompt: The planet closest to the Sun is
Target new: Earth
Before edit: Earth, which is about 1.5 times the size of the Sun.

The planet's
Before hit target: True
After edit: Earth, the planet's Earth-centered Earth. If you've ever wondered what eating a vegetable or
After hit target: True
EasyEdit rewrite_acc: [1.0] -> [1.0]

Final Summary

Total samples: 10
Successful edits: 10/10
Success rate: 100.00%

指标	成功数量 / 总数	百分比
ES / Efficacy	10 / 10	100%
PS / Generalization	7 / 10	70%
NS / Locality	6 / 10	60%

Task 3: 批量知识编辑实践 (Batch Editing using MEMIT)

由于实验设备显存和磁盘资源有限。原始 500 条 MEMIT 批量编辑在本机上存在较大的资源压力。因此先选取 CounterFact 中的 50 条样本完成 MEMIT 批量编辑实验。

```
(easyedit) D:\LLM\python_run_memit_500.py
2023-12-17 17:50:03.023130 17940 site-packages\torch\distributed\elastic\multiprocessing\redirects.py:29] NOTE: Redirects are currently not supported in Windows or MacOS.

=====
MEMIT Batch Editing - gpt2-medium
=====
Dataset: D:\LLM\counterfact_500_easyedit.json
Number of edits: 50
Hparams: D:\LLM\EasyEdit\hparams\MEMIT\gpt2-medium.yaml
=====

Loading model and MEMIT hparams...
Model loaded.
GPU memory before editing: 460 MB

Running MEMIT batch editing...
05/12/2026 17:50:21 - WARNING - datasets.builder - Reusing dataset wikipedia (D:\LLM\hf_cache\datasets\wikipedia\20200501.en\1.0.0\009f923d966dd00c00c8dc7f408f2b47745d44f5b7982a21f9448f4afbe475)
05/12/2026 17:51:05 - WARNING - datasets.builder - Reusing dataset wikipedia (D:\LLM\hf_cache\datasets\wikipedia\20200501.en\1.0.0\009f923d966dd00c00c8dc7f408f2b47745d44f5b7982a21f9448f4afbe475)
05/12/2026 17:51:32 - WARNING - datasets.builder - Reusing dataset wikipedia (D:\LLM\hf_cache\datasets\wikipedia\20200501.en\1.0.0\009f923d966dd00c00c8dc7f408f2b47745d44f5b7982a21f9448f4afbe475)
05/12/2026 17:52:46 - WARNING - datasets.builder - Reusing dataset wikipedia (D:\LLM\hf_cache\datasets\wikipedia\20200501.en\1.0.0\009f923d966dd00c00c8dc7f408f2b47745d44f5b7982a21f9448f4afbe475)
05/12/2026 17:53:45 - WARNING - datasets.builder - Reusing dataset wikipedia (D:\LLM\hf_cache\datasets\wikipedia\20200501.en\1.0.0\009f923d966dd00c00c8dc7f408f2b47745d44f5b7982a21f9448f4afbe475)

MEMIT editing finished.
Elapsed time: 585.84 seconds
GPU memory after editing: 3069 MB
Peak GPU memory by nvidia-smi: 3922 MB
Torch peak allocated: 2438.87158203125 MB
Torch peak reserved: 2654.0 MB

Validating first 20 edited facts...
Validation success: 16/20

Saved result to: D:\LLM\memit_500_results.json
Saved GPU log to: D:\LLM\memit_gpu_log.csv

=====
Final Summary
=====
Algorithm: MEMIT
Model: gpt2-medium
Batch edits: 50
Elapsed time: 585.84 seconds
Peak GPU memory: 3922 MB
Torch peak allocated: 2438.87158203125 MB
Validation success on first 20: 16/20
Error: None
=====
```

Elapsed time: 585.84 seconds

GPU memory after editing: 3069 MB

Peak GPU memory by nvidia-smi: 3922 MB

Torch peak allocated: 2438.87158203125 MB

Torch peak reserved: 2654.0 MB

Validating first 20 edited facts...

Validation success: 16/20

Saved result to: D:\LLM\memit_500_results.json

Saved GPU log to: D:\LLM\memit_gpu_log.csv

=====

Final Summary

Algorithm: MEMIT

Model: gpt2-medium

Batch edits: 50

Elapsed time: 585.84 seconds

Peak GPU memory: 3922 MB

Torch peak allocated: 2438.87158203125 MB

Validation success on first 20: 16/20

Task 4: 综合评估 (Comprehensive Evaluation)

```
(easymedit) D:\LLM>python evaluate.py
=====
Comprehensive Evaluation Results
=====
Algorithm: MEMIT
Model: gpt2-medium
Batch edits: 50
Validation samples: 20
=====
ES / Efficacy:      16/20 = 80.00%
PS / Generalization: 11/20 = 55.00%
NS / Locality:      20/20 = 100.00%
=====
Elapsed time: 333.63656401634216 seconds
Peak GPU memory: 3785 MB
Torch peak allocated: 1954.38671875 MB
Torch peak reserved: 2246.0 MB
Error: None
```

Task 4 中的评估指标数据		
指标	成功数 / 验证数	百分比
ES / Efficacy	16 / 20	80.00%
PS / Generalization	11 / 20	55.00%
NS / Locality	20 / 20	100.00%

MEMIT 批量编辑效果分析

本实验使用 MEMIT 算法对 gpt2-medium 进行批量知识编辑。实验从 CounterFact 数据集中选取 50 条样本作为批量编辑集，并在编辑完成后选取前 20 条样本进行综合评估。评价指标包括编辑成功率 ES、泛化性 PS 和局部性 NS。最终结果显示：ES 为 80.00%，PS 为 55.00%，NS 为 100.00%。也就是说，在直接编辑 Prompt 上，20 条验证样本中有 16 条能够输出目标答案；在同义改写 Prompt 上，有 11 条能够输出目标答案；在无关事实测试中，20 条样本均未出现明显破坏。

从结果看，MEMIT 在直接编辑 Prompt 上表现较好，说明它能够将一批新知识注入模型，并使模型在原始编辑提示下较稳定地输出目标答案。例如，第 1 条样本将 “Angola is located in” 的目标答案编辑为 “Antarctica”，编辑后模型输出 “Angola is located in Antarctica...”，成功命中目标答案。第 5 条样本将 “Google Patents” 的开发者编辑为 “Microsoft”，编辑后模型输出 “Google Patents, a product developed by Microsoft...”，也成功命中目标答案。

但是，PS 只有 55.00%，说明 MEMIT 的编辑效果并不能完全泛化到同义改写 Prompt。也就是说，模型在原始 Prompt 上可能已经学会输出目标答案，但当问题换一种表达方式时，模型仍然可能回到旧知识、输出无关内容，或者受到改写句子中上下文噪声的干扰。这说明 MEMIT 的批量编辑更擅长在“编辑时见

过的 Prompt 模式”上生效，而对语义等价但表面形式不同的输入，泛化能力仍然有限。

局部性 NS 达到 100.00%，说明在本次验证的 20 条样本中，模型对无关事实的回答没有被明显破坏。需要注意的是，本实验的 CounterFact 转换数据中没有 locality_ground_truth 字段，因此 NS 使用的是“编辑前后 locality prompt 输出是否保持相近”的近似计算方式，而不是严格判断无关事实是否等于标准答案。这个结果可以说明本次 MEMIT 编辑在验证样本上没有明显扰动无关输出，但不能完全证明模型所有无关知识都保持正确。

失败案例总结

失败案例 1：直接编辑失败

第 2 条样本的编辑目标是：

Prompt: What is the twin city of Shanghai? It is

Target new: Dresden

但是编辑后模型输出：

the capital of China, and the largest city in

没有输出目标答案 Dresden，因此 ES 失败。该失败说明，对于某些关系表达较复杂或模型本身理解不稳定的 Prompt，MEMIT 即使完成了批量编辑，也不一定能让模型在直接 Prompt 上输出目标对象。

失败案例 2：原始 Prompt 仍然回到旧知识

第 3 条样本的目标是将：

2011 Cannes Film Festival can be found in

编辑为：

Prescott

但编辑后模型输出：

the beautiful French Riviera town of Cannes

这里模型仍然输出了与旧知识相关的 Cannes，没有命中目标答案 Prescott。这类失败说明，当模型对某些实体和事实具有较强的原始记忆时，MEMIT 的编辑可能不足以覆盖模型原有知识。

失败案例 3：同义改写泛化失败

第 1 条样本在直接 Prompt 上编辑成功：

Angola is located in -> Antarctica

编辑后模型确实输出了 Antarctica。但是在同义改写 Prompt：

Plaque - located at Punkeydoodle's Corners. Angola belongs to the continent of

模型输出的是：

Africa

没有输出目标答案 Antarctica，因此 PS 失败。这个例子说明，虽然 MEMIT 能够让模型在直接编辑 Prompt 上输出目标答案，但当输入形式变复杂、带有额外上下文噪声时，编辑效果可能无法泛化。

失败案例 4：同义改写受到无关上下文干扰

第 6 条样本的目标是将 Lady Gaga 的相关答案编辑为 violin。直接 Prompt 上模型输出了：violinist

因此 ES 成功。但在 rephrase prompt 中，模型输出：

role of the mother of the child

没有命中 violin，导致 PS 失败。该样本说明，CounterFact 的 rephrase prompt 中往往包含额外的上下文片段，这些上下文可能会干扰模型生成，使模型没有继续围绕目标事实作答。

失败案例 5：长实体或复杂实体编辑失败

第 20 条样本的 subject 是：

Elizabeth Boleyn, Countess of Wiltshire

目标答案是：

Egypt

但编辑后模型输出：

the village of Wiltshire, England

没有输出 Egypt，ES 失败。这个失败可能与 subject 较长、实体边界复杂有关。MEMIT 和 ROME 类方法都需要定位 subject 表示，长实体或带有头衔、逗号的复杂实体可能导致定位不稳定，从而影响编辑效果。